

Cancer is a broad group of diseases characterized by the uncontrolled growth, division, and survival of abnormal cells in the body. Under normal conditions, cells grow, divide, and die in a controlled manner through a process known as the cell cycle. Cancer develops when genetic mutations disrupt this regulation, allowing cells to multiply uncontrollably and evade programmed cell death, also called apoptosis. These abnormal cells can accumulate to form masses known as tumors. Tumors may be benign, meaning they do not spread to other parts of the body, or malignant, meaning they invade nearby tissues and can spread to distant organs through the blood or lymphatic system, a process known as metastasis.

The primary cause of cancer is damage to DNA within cells. This damage may be inherited from parents or acquired during a person's lifetime due to environmental and lifestyle factors. Common causes include tobacco use, which is responsible for a significant proportion of lung and other cancers, excessive alcohol consumption, unhealthy diet, obesity, lack of physical activity, exposure to ionizing radiation such as X-rays or ultraviolet radiation from sunlight, and contact with carcinogenic chemicals. Certain infections also play an important role in cancer development, including human papillomavirus in cervical cancer, hepatitis B and C viruses in liver cancer, and *Helicobacter pylori* in stomach cancer. Aging is another major risk factor, as DNA damage accumulates over time.

Cancer can affect almost any tissue or organ in the body and exists in more than one hundred different forms. Carcinomas originate in epithelial cells, which line the skin and internal organs, and include cancers such as breast, lung, prostate, and colorectal cancer. Sarcomas develop from connective tissues such as bone, muscle, fat, or cartilage. Leukemias arise in blood-forming tissues like bone marrow and result in the abnormal production of white blood cells. Lymphomas and multiple myeloma originate in the immune system, particularly lymphocytes and plasma cells. Cancers of the central nervous system affect the brain and spinal cord and can disrupt essential neurological functions.

Cancer progression is commonly described using a staging system that reflects the size of the tumor and the extent of its spread. Stage zero refers to carcinoma in situ, where abnormal cells are present but have not invaded surrounding tissue. Early stages indicate localized disease, while advanced stages signify regional spread to lymph nodes or distant metastasis to organs such as the liver, lungs, brain, or bones. Staging is critical because it guides treatment decisions and helps predict prognosis.

The symptoms of cancer vary widely depending on the type and location of the disease. General symptoms that may occur across many cancer types include unexplained weight loss, persistent fatigue, chronic pain, fever without infection, and night sweats. Local symptoms may include the presence of a lump, abnormal bleeding, changes in bowel or bladder habits, persistent cough, difficulty swallowing, or changes in skin appearance such as new moles or non-healing sores. Because many early cancers are asymptomatic, symptoms often appear only after the disease has progressed.

Cancer diagnosis involves a combination of clinical evaluation and specialized testing. Imaging techniques such as X-rays, computed tomography, magnetic resonance imaging, and positron emission tomography are used to visualize tumors and detect metastases. Laboratory tests may identify abnormal blood values or tumor markers. A definitive diagnosis usually requires a biopsy, in which a sample of tissue is examined microscopically to identify malignant cells. Advances in molecular and genetic testing now allow clinicians to analyze specific mutations and biomarkers that influence treatment selection.

Cancer treatment depends on multiple factors, including cancer type, stage, molecular characteristics, and the patient's overall health. Surgery aims to physically remove tumors and is often most effective for localized cancers. Chemotherapy uses cytotoxic drugs to kill rapidly dividing cells, but it can also damage healthy tissues, leading to side effects such as nausea, hair loss, fatigue, and immune suppression. Radiation therapy uses high-energy radiation to destroy cancer cells or shrink tumors and may be delivered externally or internally.

Immunotherapy represents a major advancement in cancer treatment by enhancing the body's immune response against cancer cells. This includes checkpoint inhibitors that remove inhibitory signals from immune cells, cancer vaccines that stimulate immune recognition, and CAR-T cell therapy, which involves genetically modifying a patient's immune cells to target cancer. Targeted therapy focuses on specific genetic mutations or signaling pathways essential for cancer cell survival, offering more precise treatment with fewer side effects than traditional chemotherapy.

Cancer prevention focuses on reducing exposure to known risk factors and promoting early detection. Preventive strategies include avoiding tobacco, limiting alcohol intake, maintaining a balanced diet rich in fruits and vegetables, engaging in regular physical activity, protecting skin from excessive sun exposure, receiving vaccinations against cancer-causing viruses, and participating in recommended screening programs. Screening methods such as mammography, Pap smears, colonoscopy, and low-dose CT scans can detect cancer at an early, more treatable stage.

The prognosis of cancer varies greatly depending on cancer type, stage at diagnosis, biological behavior, and response to treatment. Advances in early detection, precision medicine, and supportive care have significantly improved survival rates for many cancers. In cases where cure is not possible, palliative care plays an essential role by focusing on symptom relief, pain management, and psychological support to improve quality of life for patients and their families.

Ongoing cancer research aims to improve understanding of tumor biology, identify new therapeutic targets, develop personalized treatment approaches, and integrate artificial intelligence into diagnosis and treatment planning. These efforts continue to transform cancer from a uniformly fatal disease into a manageable or curable condition for many patients.